A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 15

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (2900 - 1400) / 1400 x 100 **1** = 107% (accept 107.1) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Large decrease from 24 per year (1975-79) to 2 per year (2010-14) **1** Steepest fall between the late 1970s and 1980s – slower decline / levelling off after 1995 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Double-hulled tankers required for all large tankers **1** Improved navigation (GPS, AIS) and offshore shipping routes **1** Inert gas systems reducing explosions **1** Twin engines / rudders reducing groundings after mechanical failure **1** Stricter regulation, inspection and polluter-pays liability (eg after Exxon Valdez) **1** Separate ballast tanks / crude oil washing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
01.4	**1 mark each** 1 mark for each of the 3 points below. **1** Most oil pollution comes from dispersed / diffuse sources not tanker accidents: waste lubricating oil, industrial oil in drains, road runoff **1 Routine operations:** small spills, pipeline leaks, storage tank leakage, oil rig accidents (eg Deepwater Horizon) **1** Growth in oil use and transport means more small releases even if large accidents are rarer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **diffuse** → pollution from a wide area with no single outlet	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-262; AQA spec 3.4.3.2.7.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Many different organic substances may be present, each with a different potential to cause deoxygenation, so it is difficult to measure or forecast them individually **1** BOD measures the effect (amount of oxygen used by microbes decomposing the organic matter) under standardised conditions (eg 5 days, 20 °C, dark) **1** It gives a single measure directly related to the risk of deoxygenation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Biochemical Oxygen Demand** → the oxygen used by aerobic decomposers in a water sample • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'a high BOD means the water contains a lot of oxygen' - a high BOD means a lot of oxygen will be removed	3

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** E. coli is a common gut bacterium so its presence confirms faecal contamination and its number indicates how serious the contamination is **1** Pathogens are not always present so their absence would not prove the water is uncontaminated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Different species have different sensitivities to pollution **1** Easy to identify **1** Easy to find / normally present / usually common **1** Widely distributed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.4	**1 mark each** 1 mark for each of the 2 points below. **1 Advantage:** assesses the effect of pollution over a period of time (recent pollution events are detected even if the pollutant has gone) – no expensive equipment needed **1 Disadvantage:** identification needs skill – does not identify which pollutant is present so further chemical tests are required – species may be absent for other reasons **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 273-275; AQA spec 3.4.3.2.9.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** UV light splits diatomic oxygen (O2) into two monatomic oxygen atoms (O) **1** Monatomic oxygen reacts with diatomic oxygen to form ozone (O3) **1** UV light also splits ozone into O2 and O – the reactions form a dynamic equilibrium which absorbs UV (converting it to chemical energy / heat) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
03.2	**Marks shared** Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2 Useful:** boiling points near ambient temperature so easily liquefied under low pressure (aerosol propellants, fridge / air-conditioning coolants) – dissolve grease but do not damage electrical components (solvents) – non-flammable – non-toxic **up to 2 Persistence:** chemically stable / unreactive in the troposphere so they remain long enough to be carried up to the stratosphere **up to 2 Dissociation by UV:** in the stratosphere higher UV levels break the carbon-chlorine bonds releasing chlorine free radicals **up to 2** Chlorine reacts with monatomic oxygen (and ozone), preventing ozone formation – chlorine is regenerated so one atom destroys many ozone molecules – bromine / iodine act similarly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine • **Persistence** → does not break down / stays in the environment for a long time **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	6
03.3	**1 mark each** 1 mark for each of the 3 points below. **1** UVB energy is absorbed by cells and breaks up biological molecules / DNA causing mutations, skin cancer, cataracts, sunburn (human health) **1** Damage to leaf tissue and reduced photosynthesis / crop yields (plants) **1** Damage to marine algae, phytoplankton, corals and planktonic larvae (marine organisms), reducing productivity and food webs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
03.4	1 mark for the point below. **1** An area where the total ozone column falls below 220 Dobson Units (normal is about 300 DU) - a level never recorded before CFC pollution **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
03.5	1 mark for the point below. **1** Concentration varies with altitude and the stratosphere is not normally sampled – Dobson Units measure the total column thickness instead **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
03.6	1 mark for the point below. **1** The British Antarctic Survey at Halley detected higher UV at ground level, showing ozone above had decreased (satellites above the stratosphere cannot measure downward UV directly) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-109; AQA spec 3.2.1.3.	
04.1	1 mark for the point below. **1** The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	1
04.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Light:** UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions **1 Temperature:** higher temperatures increase the rate of chemical breakdown / microbial decomposition **1 Soil microbes / oxygen:** aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils **1** pH / presence of other chemicals may increase or decrease the rate of degradation **1** Adsorption onto clay / organic matter may protect from degradation and increase persistence **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **Adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** There is an environmental gradient (distance from the hedge / changing conditions) so samples must be taken at regular intervals along it to detect the change **1** Random sampling might leave gaps / miss the pattern – systematic sampling avoids bias while covering the gradient **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **systematic sampling** → sampling at regular intervals • **transect** → a line along which samples are taken to show change with distance • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	2
05.2	**1 mark each** 1 mark for each of the 4 points below. **1** Sample at close intervals (eg every 1 m) first and see at what spacing the variations are still detected without unnecessary data – wider intervals (eg 4 m) may miss peaks and troughs **1** Take increasing numbers of samples at one site and calculate the running mean – the number needed is where the mean stops fluctuating / stabilises **1** Calculate standard deviation to assess scatter: more variable data need more samples **1** Test the method (extraction technique, timing) for reliability before the main study **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	4

Do not write outside the box

Question	Marking guidance	Marks
05.3	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2** Remove surface vegetation – pour an irritant solution (dilute mustard / detergent in water) onto the area until saturated (and around it), keep adding as it infiltrates **up to 2** Collect worms that come to the surface for a standardised time – count / identify / weigh **up to 2 Limitations:** solution may not percolate impermeable soils or reach deep worms – worms may move sideways or downwards – some may die or not respond – slow-moving worms missed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - tiny pores between clay particles allow capillary rise – none in sand	1
06.2	**1 mark each** 1 mark for each of the 3 points below. **1 Hygroscopic:** held tightly by adhesive forces on particle surfaces – cannot drain and cannot be absorbed by plants **1 Capillary:** held in pores by surface tension – does not drain but can be absorbed by plants **1 Gravitational:** drains freely through large pores – available to plants only briefly as it passes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
06.3	1 mark for the point below. **1** Small round crumbs leave spaces giving good drainage, aeration and easy root penetration – large flat platy peds pack together reducing all three **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **peds** → soil crumbs / aggregates of soil particles	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 3 points below. **1** Trees absorb CO2 by photosynthesis and store carbon in wood as they grow **1** Once mature there is no further net storage (photosynthesis balanced by respiration / decomposition), though the standing forest remains a reservoir **1** Harvesting the timber and keeping it as wood (eg house construction) then replanting produces additional storage as the new trees grow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
07.2	**1 mark each** 1 mark for each of the 4 points below. **1 Pre-combustion:** the fuel is modified before use so no CO2 is released when burnt - eg gasification of coal to hydrogen and CO2 (CO2 stored, hydrogen distributed) – oxy-fuel combustion in pure oxygen giving only CO2 and water vapour which is condensed out **1 Advantage:** capture at a large central facility – suits small dispersed sources (vehicles) that cannot capture CO2 **1 Post-combustion:** CO2 removed from exhaust gases by solvents, membrane filtration, adsorption / desorption, cryogenic separation, graphene **1 Disadvantage:** only about 20% of flue gas is CO2 so separation is costly / energy-intensive – none fully developed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **carbon capture** → capturing carbon dioxide from flue gases and storing it underground • **adsorption** → nutrient ions sticking to the surfaces of clay and humus	4
07.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Injected into depleted oil / gas fields or saline aquifers **1** Injected into oil reservoirs to provide pressure for secondary oil recovery **1** Reaction with fractured basalt to form solid carbonate minerals **1 Concern:** leakage back to the atmosphere / monitoring over centuries – limited suitable geology – cost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 151-152; AQA spec 3.2.4.2.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** The void fills with water naturally – wetlands are uncommon / valuable habitats in most regions **1** The new habitat may have greater wildlife value than the farmland lost – low cost compared with filling and levelling **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.2	**1 mark each** 1 mark for each of the 2 points below. **1 Housing:** ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) **1 Farmland:** landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.3	1 mark for the point below. **1** Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost – transplanted plants often fail **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	
09.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** $3.6 \times 10^{11} / 1.2 \times 10^{9}$ **1** = 300 years **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (years) is already printed on the answer line. • **Residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** It must be porous, with spaces between the particles, so that it can store a large volume of water **1** It must be permeable, so that water can flow through it to a well or spring fast enough to be abstracted **1** An impermeable layer beneath it stops the water draining away, and the rock must be exposed at the surface somewhere, or overlain by permeable material, so that rainfall can recharge it **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **permeable** → how easily water passes through / lets water through • **impermeable** → does not let water through	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Abstraction of 4.8 x 10^9 m^3 per year is four times the recharge of 1.2 x 10^9 m^3 per year, so water is being removed faster than it is replaced **1** The stored water is being mined rather than used sustainably, so the saturated zone shrinks and its upper surface, the water table, falls each year **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Abstraction** → water taken out of a river, lake or aquifer	2
09.4	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Wells and boreholes must be deepened and the water pumped from further down, so the energy cost of abstraction rises and shallow wells fail first **1** Rivers, springs and wetlands fed by the aquifer dry up, destroying habitats and the species that depend on them **1** Near the coast, sea water is drawn into the aquifer (saltwater intrusion), making the water too saline for drinking or irrigation **1** The rock compacts as the water that supported it is removed, causing land subsidence that damages buildings and drainage, and permanently reduces the storage capacity **1** Irrigated agriculture that depends on the aquifer fails, with loss of food production and livelihoods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **abstraction** → water taken out of a river, lake or aquifer **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4

Do not write outside the box

Question	Marking guidance	Marks
09.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Fossil water fell as rain thousands of years ago under a wetter climate – under the present climate recharge is negligible, so any abstraction depletes a finite store and the resource is effectively non-renewable **1** Changing to crops with a lower water demand, or to drip / trickle irrigation applied at night, which delivers water to the roots and loses far less to evaporation **1** Metering and pricing water, or setting abstraction licences and quotas, so that users have a reason to waste less **1** Recycling treated waste water for irrigation or industry, and harvesting rainwater, so less is taken from the aquifer **1** Lining canals and repairing leaking pipes so that less of the water abstracted is lost before it is used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **abstraction** → water taken out of a river, lake or aquifer • **Recycling** → reprocessing a material to make something new **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 115-121; AQA spec 3.2.2.1-3.2.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Properties of renewables:** intermittency, predictability, low energy density, storage, applicability, locational constraints, impacts, technology level, economics • **Wind:** large offshore resource in shallow seas, falling costs, HAWT / VAWT / direct drive developments – intermittent, needs storage and grid – land-use conflicts • **Solar:** PV costs falling rapidly, roof space, but low winter insolation in the UK – CSP unsuitable – heat pumps and photothermal for heating • **HEP:** most large sites used – low-head, helical and micro-hydro extend use – pumped storage vital • **Tidal:** large range (Severn, Mersey), predictable – barrages costly with high impact, lagoons and in-stream turbines lower impact • **Wave:** long Atlantic fetch, but technology immature • **Biofuels:** wastes, anaerobic digestion, energy crops vs food / habitat • **Geothermal:** limited to low-temperature schemes / heat pumps • **Systems issues:** storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management, diversity of sources, energy conservation reducing demand, electrification of transport • **Judgement:** renewables could supply most electricity with storage and a diverse mix, but heat and transport are harder – transition needs investment and lifestyle change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any accurate, relevant material that is not in the indicative content. • **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous • **insolation** → incoming solar radiation / energy arriving from the Sun • **HEP** → electricity generated from falling water **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	
	Model answer - Level 3 (8 marks)	
	What renewables can contribute depends on their properties - intermittency, predictability, energy density, storage, applicability, location, impacts, technological maturity and cost - set against the UK's resources. Wind is the largest: shallow seas give a huge offshore resource, costs have fallen and direct-drive and vertical-axis designs keep improving, but it is intermittent, needs storage and grid connections and provokes land-use conflicts onshore. Solar PV costs are falling rapidly and roof space is plentiful, but winter insolation is low and concentrating solar is unsuitable; photothermal panels and heat pumps are better for heating. Most large HEP sites are used, though low-head, helical and micro-hydro extend it and pumped storage is vital for balancing. The Severn and Mersey have large, predictable tidal ranges, but barrages are costly and damaging while lagoons and in-stream turbines have lower impact; the long Atlantic fetch gives wave energy that the technology cannot yet harness reliably. Biofuels from wastes and anaerobic digestion are useful, but energy crops compete with food and habitat, and geothermal is limited to low-temperature schemes.	
	The system matters as much as the sources: storage in pumped HEP, batteries, hydrogen and power-to-gas, vehicle-to-grid, grid management, a diverse mix that smooths supply, and conservation to reduce demand. With these, renewables could supply most of the UK's electricity. Heat and transport are harder, because they need high temperatures or liquid fuels, so electrification and hydrogen would be required. The potential is large, but realising it needs sustained investment and changes in how energy is used.	
	Weaker answer - Level 1 (3 marks)	
	The UK could use lots of renewable energy. It is windy so wind turbines could be built, especially offshore. Solar panels can be put on roofs but it is not very sunny in winter. Tidal power could be used in the Severn estuary because it has big tides. Hydroelectric power can be used in Scotland. Renewables are intermittent so batteries would be needed. Renewables could provide a lot of the UK's energy in the future.	
	What makes the difference	
	• Level 1 mentions the main resources with one fact each; Level 3 assesses each against its properties, constraints and the state of the technology.	
	• Level 3 treats storage, grid and demand as a system and distinguishes electricity from heat and transport - essential for judging 'future energy supplies'.	

Do not write outside the box

Question	Marking guidance	Marks
	• The conclusion in Level 3 is qualified (most electricity; heat and transport harder; needs investment); Level 1's is unqualified.	
10.2	1 mark for the point below. **1** Eight times (energy proportional to velocity cubed) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-217; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • **Renewable features:** intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal) – low energy density – location – low pollution but manufacturing impacts • Technologies and developments for each resource with examples – new technologies improving usability • **Storage and grid management:** pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving – the role of electricity as a secondary fuel • **Economics and politics:** development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution – sustainability principles - natural systems run on low-density renewable energy • **Judgement:** electricity largely yes	25
	– heat, aviation, shipping and high-temperature industry harder – transition period with nuclear / CCS *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Features of fertile soil:** water, nutrients, air, organic matter, pH, biota, texture, structure, depth – slow rate of soil formation vs rapid erosion (dynamic equilibrium broken) • **Causes of degradation:** vegetation removal, ploughing vulnerable soils, overgrazing, reduced organic matter and biota, cultivating steep slopes, compaction, salinisation from irrigation, nutrient depletion, pollution • **Types of erosion:** rain splash, runoff, wind, slumping – consequences: reduced productivity, river and reservoir sedimentation, flooding, coastal sedimentation, particulates, desertification, landslides • USLE as a tool for estimating loss and testing management • **Management:** long-term crops, low / zero tillage, contour ploughing, tied ridging, terracing, stone rows, windbreaks, multicropping / strip cropping, mulching, organic matter, livestock control, drainage / pH / nutrient management, conservation of soil biota • Links to sustainable agriculture, carbon storage, hydrological cycle, circular economy	25
	• Evaluation and judgement on which strategies are most effective in different environments *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169, 328-337; AQA spec 3.2.5, 3.5.1.3, 3.5.1.5.	

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